

Department of Mathematics and Natural Sciences

Mid Term Examination

Semester: Fall 2025

Course Title: MATHEMATICS I: Differential Calculus & Coordinate Geometry

Course Code: MAT-110

Time: 1 hour 20 minutes

Date: November 23, 2025

Total Marks: 40. It will be converted into 20.

SOLUTION & MARKING SCHEME

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NOTE:

TO APPRECIATE YOUR MATHEMATICAL THINKING, ANY VALID MATHEMATICAL WAY OF SOLVING WILL BE TREATED EQUALLY AND ASSESSED ACCORDINGLY, AS LONG AS IT'S CONSISTENT WITH THE PROVIDED INSTRUCTIONS OF THE QUESTION.

IN THIS ASSESSMENT PROCESS EVERY PROBLEM WILL BE ASSESSED BASED ON 3 CRITERIA:

- A. MATHEMATICAL THINKING
- B. MATHEMATICAL KNOWLEDGE
- C. MATHEMATICAL SKILL

PERCENTAGE SCHEME

- N/A = Not Assessed / Not Answered
 - 0% = Incorrect
 - 50% = Partially Correct
 - 80% = Mostly Correct
 - 100% = Perfect
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QUESTION_1:

Define continuity. Determine whether $g(t)$ is continuous at $t = 1$, where

$$g(t) = \begin{cases} \frac{1}{t}, & t < 0, \\ t^2 + 1, & 0 \leq t \leq 1, \\ 2t, & t > 1 \end{cases}$$

Solution:

Definition of Continuity

A function $f(x)$ is said to be **continuous** at $x = a$ if:

1. $f(a)$ exists
 2. $\lim_{x \rightarrow a} f(x)$ exists
 3. $\lim_{x \rightarrow a} f(x) = f(a)$
-

Step 1: Value of the function at $t = 1$

Since $0 \leq t \leq 1$,

$$g(1) = 1^2 + 1 = 2$$

Step 2: Left-hand limit

$$\lim_{t \rightarrow 1^-} g(t) = \lim_{t \rightarrow 1^-} (t^2 + 1) = 2$$

Step 3: Right-hand limit

$$\lim_{t \rightarrow 1^+} g(t) = \lim_{t \rightarrow 1^+} (2t) = 2$$

Step 4: Comparison

$$\lim_{t \rightarrow 1^-} g(t) = \lim_{t \rightarrow 1^+} g(t) = g(1) = 2$$

Conclusion

The function $g(t)$ is continuous at $t = 1$.

QUESTION_2:

Find $\frac{dy}{dx}$ of the function

$$y = 2 \sin^{-1} \left(\frac{5x}{2x^2 + 1} \right)$$

Solution:

Let,

$$u = \frac{5x}{2x^2 + 1}$$

Step 1: Apply chain rule

$$\frac{dy}{dx} = 2 \cdot \frac{1}{\sqrt{1 - u^2}} \cdot \frac{du}{dx}$$

Step 2: Differentiate u (quotient rule)

$$\frac{du}{dx} = \frac{5(2x^2 + 1) - 5x(4x)}{(2x^2 + 1)^2} = \frac{5 - 10x^2}{(2x^2 + 1)^2}$$

Step 3: Evaluate $1 - u^2$

$$1 - u^2 = \frac{(2x^2 + 1)^2 - 25x^2}{(2x^2 + 1)^2}$$

Final Answer

$\frac{dy}{dx} = \frac{2(5 - 10x^2)}{(2x^2 + 1)\sqrt{(2x^2 - 5x + 1)(2x^2 + 5x + 1)}}$
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QUESTION_3:

Find the n -th order derivative of $y = (3x + 5)^m$, if $x \neq -\frac{5}{3}$ and $m > n$.

Hence find the n -th derivative if $m = n$.

Solution:

(a) Case: $m > n$

Each differentiation produces a factor of 3.

$$\frac{d^n y}{dx^n} = 3^n m(m-1)(m-2) \cdots (m-n+1)(3x+5)^{m-n}$$

(b) Case: $m = n$

$$\frac{d^n y}{dx^n} = 3^n n!$$

QUESTION_4:

(a) Define critical points

Solution:

A **critical point** of a function $f(x)$ is a point where:

- $f'(x) = 0$, or
 - $f'(x)$ does not exist
-

(b) Given $f(x) = 2x^3 + 3x^2 - 36x$, find intervals of increase and decrease

Step 1: First derivative

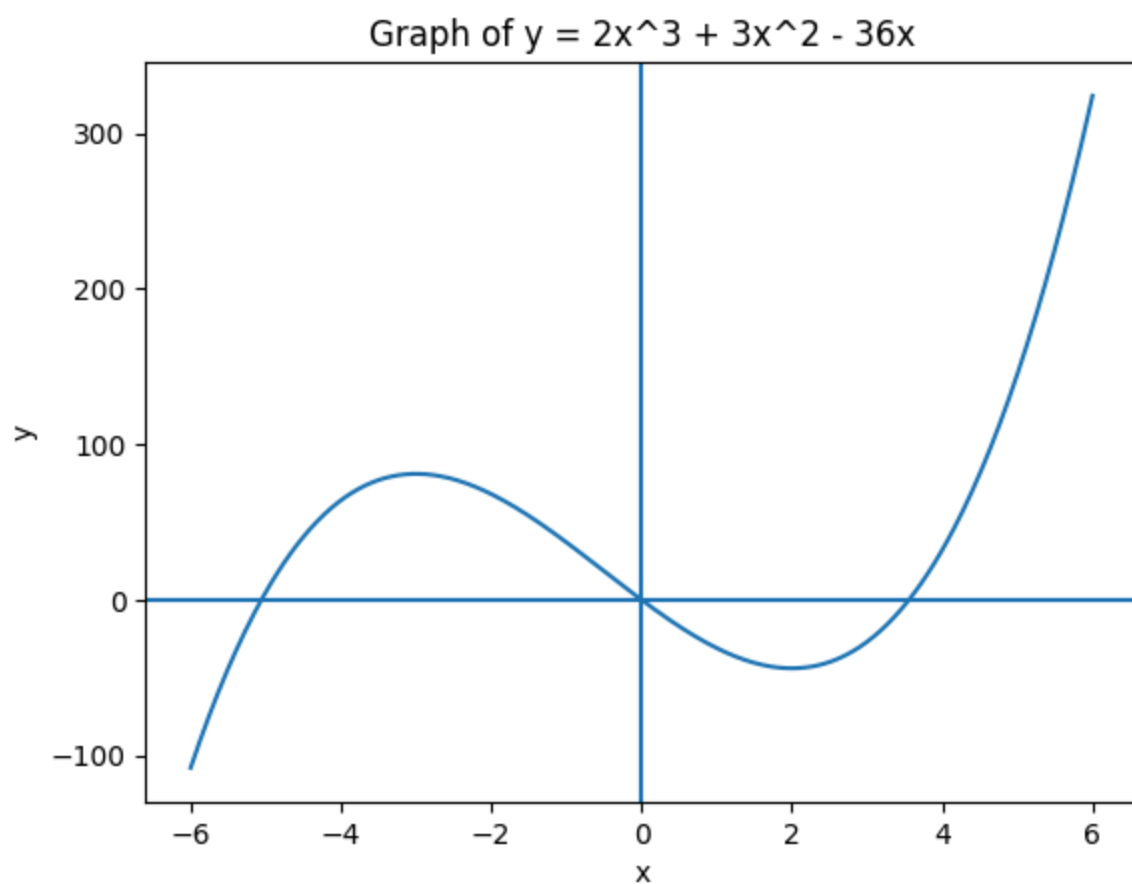
$$f'(x) = 6x^2 + 6x - 36 = 6(x + 3)(x - 2)$$

Step 2: Critical points

$$x = -3, 2$$

Step 3: Sign chart

Interval	Nature
$(-\infty, -3)$	Increasing
$(-3, 2)$	Decreasing
$(2, \infty)$	Increasing



First Derivative Test (already done)

- Relative **maximum** at $x = -3$
 - Relative **minimum** at $x = 2$
-

Second Derivative Test (Additional Approach)

$$f''(x) = \frac{d}{dx}(6x^2 + 6x - 36) = 12x + 6$$

At $x = -3$:

$$f''(-3) = 12(-3) + 6 = -30 < 0$$

$\Rightarrow$ Relative maximum at $x = -3$

At $x = 2$:

$$f''(2) = 12(2) + 6 = 30 > 0$$

$\Rightarrow$ Relative minimum at $x = 2$

Conclusion

- Relative maximum at $(-3, 81)$
- Relative minimum at $(2, -44)$

Graph of $y = f(x)$

The plotted graph (shown above) confirms:

- A **peak** near $x = -3$
- A **valley** near $x = 2$

✓ Hence, analytical and graphical results agree.

QUESTION_5:

State Rolle's theorem. Verify it for

$$f(x) = (x^2 - 2x)e^x \text{ on } [0, 2].$$

Solution:

Rolle's Theorem

If $f(x)$ is continuous on $[a, b]$, differentiable on (a, b) , and $f(a) = f(b)$, then

$$\exists c \in (a, b) \text{ such that } f'(c) = 0$$

Verification

$$f(0) = f(2) = 0$$

Given:

$$f(x) = (x^2 - 2x)e^x$$

Derivative

Using product rule:

$$\begin{aligned} f'(x) &= (2x - 2)e^x + (x^2 - 2x)e^x \\ &= e^x(x^2 - 2) \end{aligned}$$

Solve $f'(x) = 0$

$$e^x(x^2 - 2) = 0$$

Since $e^x \neq 0$ for all x ,

$$x^2 - 2 = 0 \Rightarrow x = \pm\sqrt{2}$$

Check interval $(0, 2)$

$$-\sqrt{2} \notin (0, 2), \quad \sqrt{2} \in (0, 2)$$

Conclusion

$$\boxed{c = \sqrt{2}}$$

Thus, Rolle's theorem is verified.

QUESTION_6:

Find the n -th Maclaurin polynomial of

$$f(x) = e^{2x+3}.$$

Solution:

Maclaurin's Formula

For a function $f(x)$,

$$f(x) = f(0) + xf'(0) + \frac{x^2}{2!}f''(0) + \frac{x^3}{3!}f'''(0) + \cdots$$

Given Function

$$f(x) = e^{2x+3}$$

Derivatives

$$f(0) = e^3$$

$$f'(x) = 2e^{2x+3} \Rightarrow f'(0) = 2e^3$$

$$f''(x) = 4e^{2x+3} \Rightarrow f''(0) = 4e^3$$

$$f'''(x) = 8e^{2x+3} \Rightarrow f'''(0) = 8e^3$$

Maclaurin Expansion (up to 3rd order)

$$f(x) = e^3 + x(2e^3) + \frac{x^2}{2!}(4e^3) + \frac{x^3}{3!}(8e^3) + \dots$$

$$= e^3 \left(1 + 2x + 2x^2 + \frac{4}{3}x^3 + \dots \right)$$

Final Answer

$$f(x) = e^3 \left(1 + 2x + 2x^2 + \frac{4}{3}x^3 + \dots \right)$$
